SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA0408 **COURSE NAME: Operating Systems**

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* *(BL2, BL3)*

QUESTIONS: 5 **Total Marks:100**

Annexure A

Concept Mapping

Code of Conduct:

*I __**Dharshini/192521408**__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dharshini

Annexure A

Concept Mapping

Question 1
Show a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation). **(BL2)**

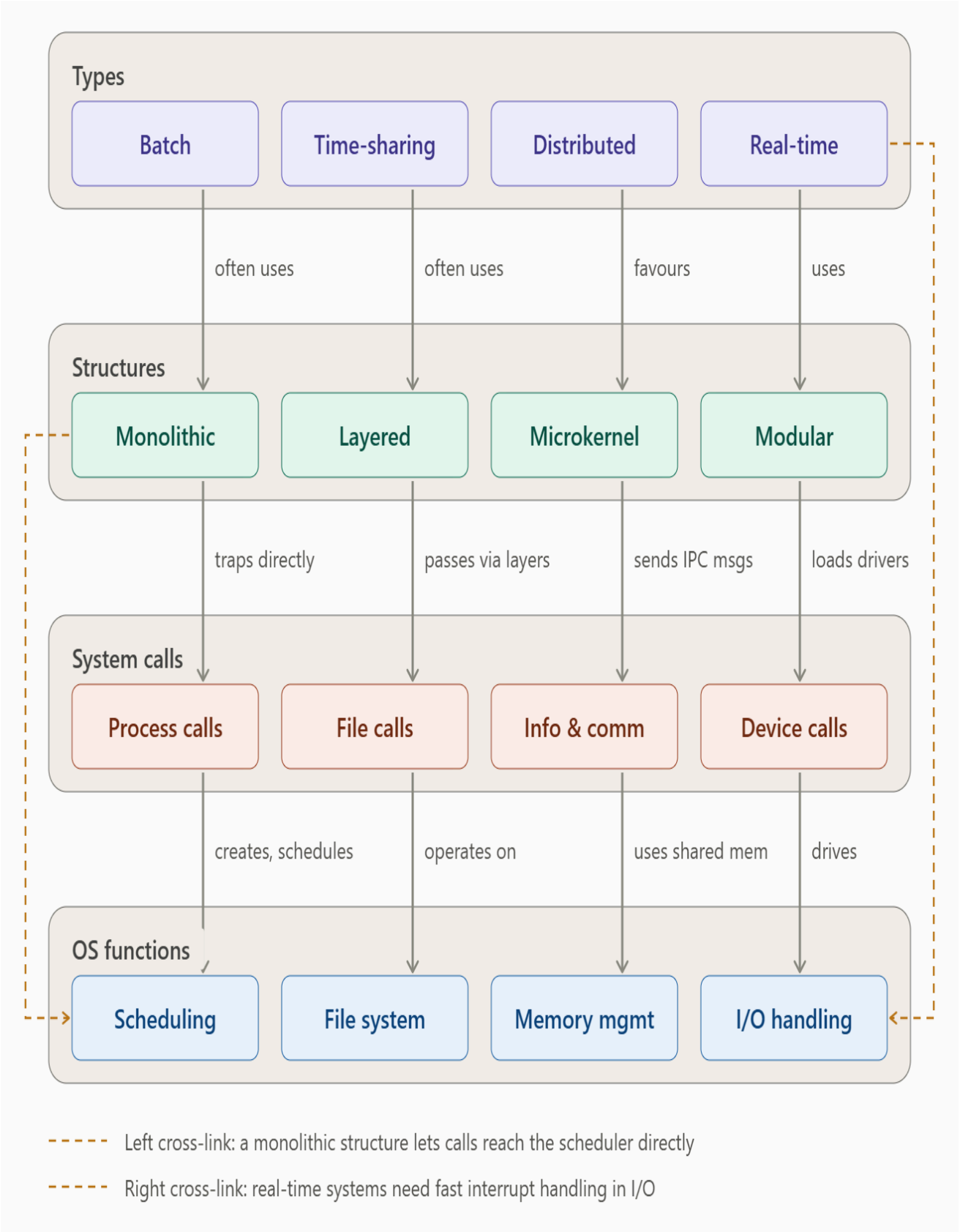
Question 2
Construct a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component. **(BL3)**

Question 3
Sketch a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases. **(BL2)**

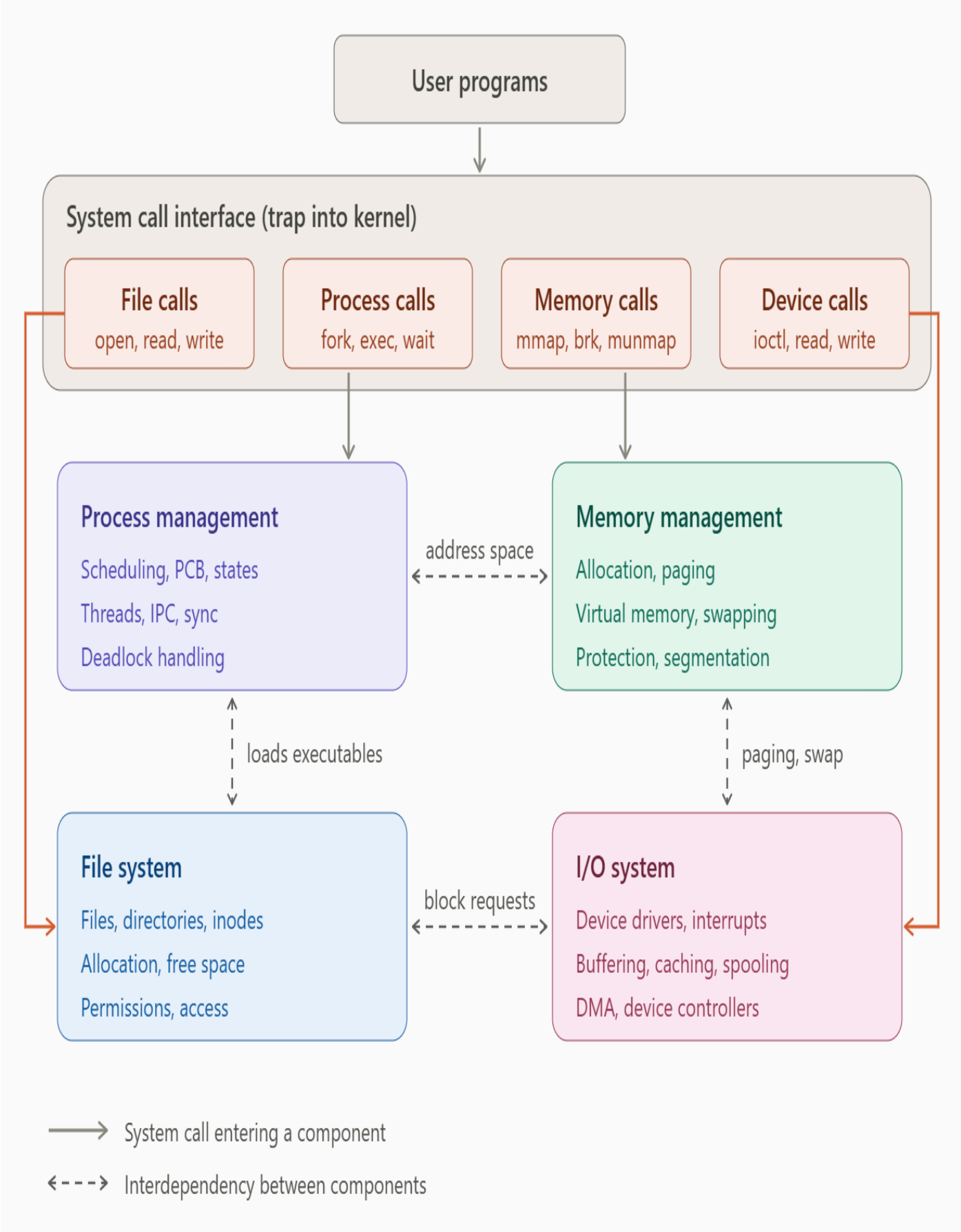
Question 4
Illustrate a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths. **(BL3)**

Question 5
Sketch a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling. **(BL3)**

Answer1:


Types
Batch
Time-sharing
Distributed
Real-time
often uses
often uses
favours
uses
Structures
Monolithic
Layered
Microkernel
Modular
traps directly
passes via layers
sends IPC msgs
loads drivers
System calls
Process calls
File calls
Info & comm
Device calls
creates, schedules
operates on
uses shared mem
drives
OS functions
Scheduling
File system
Memory mgmt
I/O handling
Left cross-link: a monolithic structure lets calls reach the scheduler directly
Right cross-link: real-time systems need fast interrupt handling in I/O

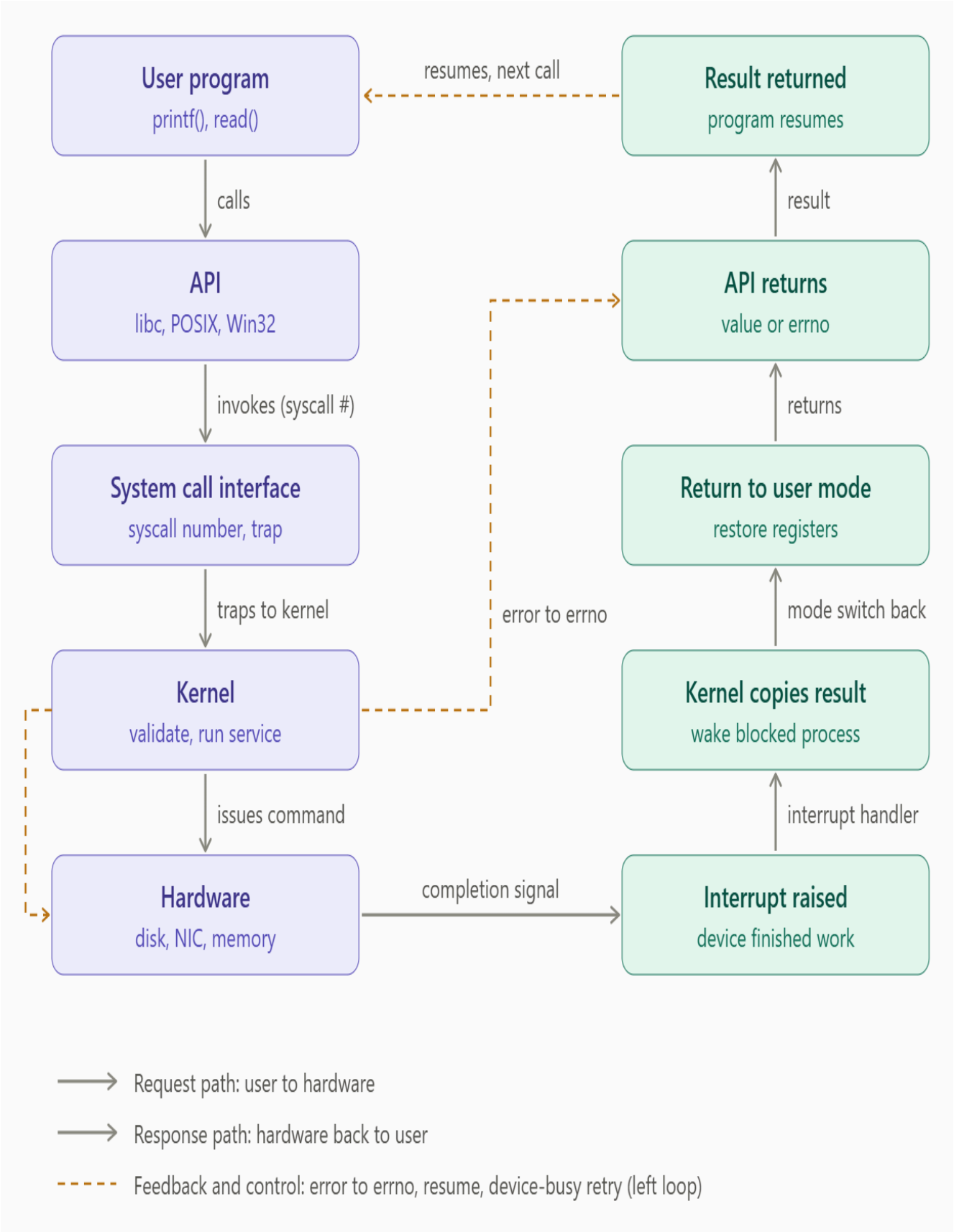
Answer2:


User programs
System call interface (trap into kernel)
File calls
open, read, write
Process calls
fork, exec, wait
Memory calls
mmap, brk, munmap
Device calls
ioctl, read, write
Process management
Scheduling, PCB, states
Threads, IPC, sync
Deadlock handling
address space
Memory management
Allocation, paging
Virtual memory, swapping
Protection, segmentation
loads executables
paging, swap
File system
Files, directories, inodes
Allocation, free space
Permissions, access
block requests
I/O system
Device drivers, interrupts
Buffering, caching, spooling
DMA, device controllers
System call entering a component
Interdependency between components

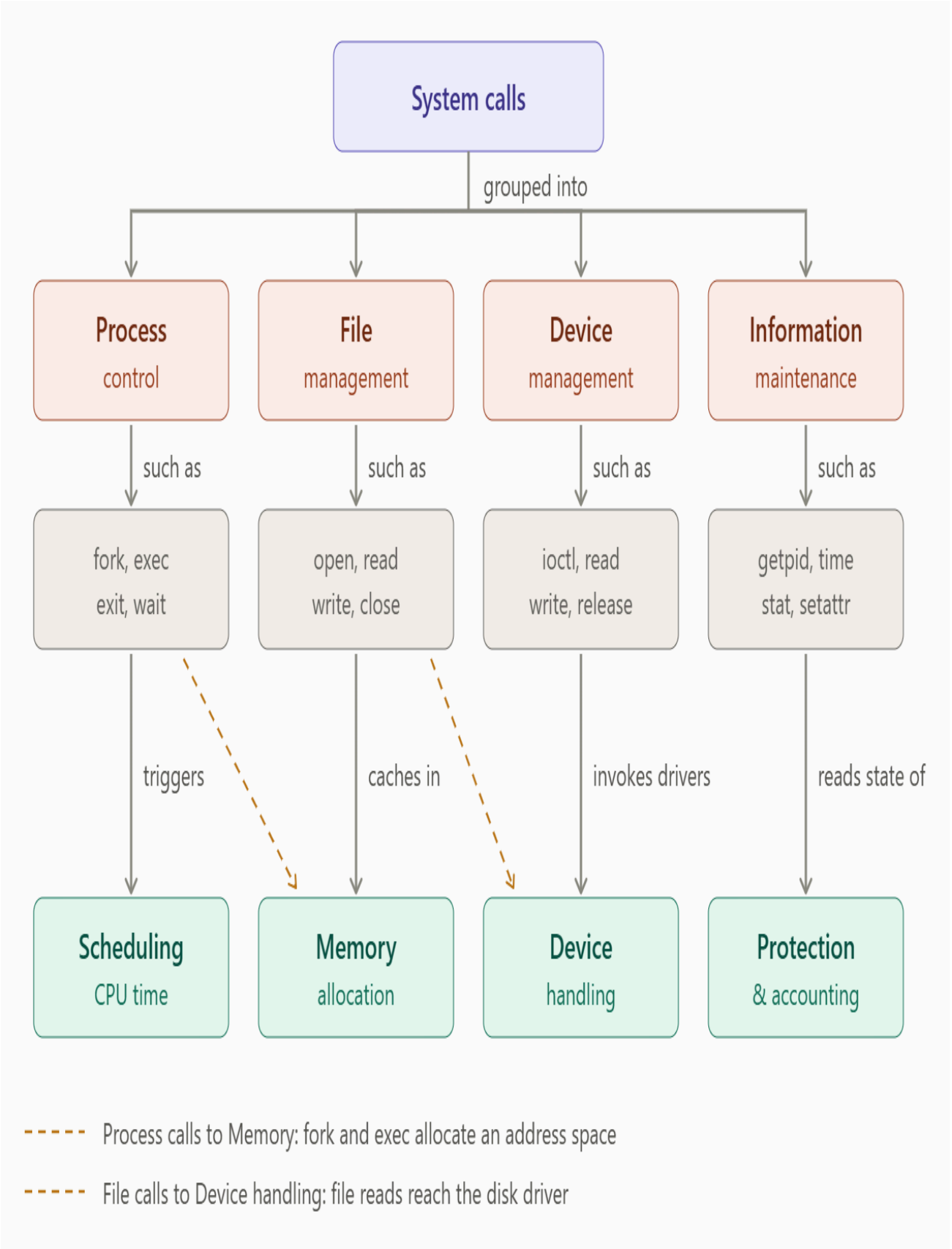
Answer3:


Operating systems
classified as
Batch
Time-sharing
Distributed
Real-time
aims for
aims for
aims for
aims for
High throughput
Fast response
Resource sharing
Meet deadlines
gains
gains
gains
gains
Efficient bulk work
Interactive, fair
Scalable, resilient
Predictable timing
suffers from
suffers from
suffers from
suffers from
No interaction
Context-switch cost
Network complexity
Rigid, costly
such as
such as
such as
such as
IBM OS/360
Unix, Linux
Plan 9, Amoeba
VxWorks, FreeRTOS
all share
all share
all share
all share
Similarities: CPU, memory, file and I/O management via system calls

Answer4:


User program
printf(), read()
resumes, next call
Result returned
program resumes
calls
result
API
libc, POSIX, Win32
API returns
value or errno
invokes (syscall #)
returns
System call interface
syscall number, trap
Return to user mode
restore registers
traps to kernel
error to errno
mode switch back
Kernel
validate, run service
Kernel copies result
wake blocked process
issues command
interrupt handler
Hardware
disk, NIC, memory
completion signal
Interrupt raised
device finished work
Request path: user to hardware
Response path: hardware back to user
Feedback and control: error to errno, resume, device-busy retry (left loop)

Answer5:


System calls
grouped into
Process
control
File
management
Device
management
Information
maintenance
such as
such as
such as
such as
fork, exec
exit, wait
open, read
write, close
ioctl, read
write, release
getpid, time
stat, setattr
triggers
caches in
invokes drivers
reads state of
Scheduling
CPU time
Memory
allocation
Device
handling
Protection
& accounting
Process calls to Memory: fork and exec allocate an address space
File calls to Device handling: file reads reach the disk driver

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand